1. **Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system? [3 points]**

It's important to push for transparency in ML/AI systems because many modern models, especially deep learning, work like black boxes. It makes it challenging for people to understand how decisions are made. Explainability makes it clear to users how the system works. It makes them more likely to trust its results, especially in high-stakes fields like healthcare, finance, or public services. For businesses, transparency helps them follow the rules. It makes it easier to find and fix bias and builds trust in the system's strength by showing which inputs really affect predictions. It also helps developers train models better by showing where they are wrong in their reasoning. Overall, explainability is an important part of responsible AI because it helps build trust, fairness, and responsibility.

2. **What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like. [3 points]**

A fair system doesn't treat people or groups unfairly because of things like race, gender, age, or disability. Instead, it makes decisions based on important, relevant factors. For instance, in an AI loan-approval model, fairness means that two people with similar financial situations would get the same answer. Independent of their race or gender was. To be fair, the model would have to use the right financial variables instead of hidden proxies like ZIP code or patterns in the data that show past discrimination. It also means giving applicants reasons for the decision. This is so that they understand why it was made and contest mistakes. Regular checks and audits would help make sure that no one group is always treated worse than another.

3. **How do you see concepts like bias, fairness, accountability, and transparency-related? [2 points]**

Bias, fairness, accountability, and transparency are all important parts of making sure that an AI system acts responsibly. Bias is when data or predictions are wrong in some way. The goal of getting rid of or lowering those harmful distortions. It is to make sure that all groups are treated fairly. People can see where bias might be happening in the model's decision-making because it is clear. When bias or unfairness is found, accountability makes sure that someone is responsible for fixing it. To put it another way, transparency shows the problem, bias is the problem, fairness is the goal, and accountability is the way to make things better.

4. **Give an example of a system that is biased but fair. [2 points]**

A system can be statistically biased and fair if the bias affects all groups equally and doesn't cause too much harm. For instance, a model that predicts medical risks might always overestimate everyone's risk by a marginal amount ei. five percent. This is technically bias because the predictions don't match the real values. However, the system could still be seen as fair because the bias is the same for all demographic groups. As a result, no one group is treated worse or denied opportunities. In some real-life situations, this kind of uniform bias is even planned. It is especially biased in healthcare where conservative estimates can make things safer without being unfair.